



Probe and analogue: Double roles of thionine for aloe-emodin selective and sensitive ratiometric detection

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ABSTRACT

Molecular imprinted polymer (MIP) has been employed for electrochemical sensor preparation with good selectivity. However, to restrict the non-specific adsorption, the conductivity of MIP is usually unsatisfying, which limits its sensitivity for template detection. To achieve both of selectivity and sensitivity, a MIP electrochemical sensor was designed and fabricated by the electropolymerization of pyrrole in the presence of aloe-emodin (AE, template molecule) on carbon nanoparticles modified electrode in this work. Thionine (TH), being added in solution in advance, played double roles for AE recognition and detection. As an analogue of AE, TH was adsorbed into imprinted cavities and then was replaced by AE, indicating the good selectivity of the sensor. This replacement of TH by AE was also demonstrated by the result of computational simulation. As an electro-active probe, the peak current of TH decreased and peak current of AE increased based on the replacement of TH by AE. A current ratiometric signal defined as $|\Delta i_{AE}|/|\Delta i_{TH}|$ was used for AE sensitive detection. Thus, the replacements of TH by AE demonstrated that the sensor possessed high sensitivity and good selectivity. A linear range of 1.0×10^{-7} – 1.0×10^{-4} mol/L for AE detection was obtained with a limit of detection of 7.5×10^{-8} mol/L. And the sensor has been applied to analyze AE in real samples with satisfactory results. The simple, smart, and effective strategy presented in this work can be further used to prepare electrochemical sensors for other compounds selective recognition and sensitive detection.

1. Introduction

Development of molecularly recognized materials has attracted much attention because of their significant contribution in the fields of solid phase extraction [1,2], chromatography separation [3], biomimetic catalysis [4], and chemical sensing [5–8]. Molecularly imprinted polymer (MIP), as an artificially synthesized receptor, has large amount of imprinted cavities with the complementary shape, size, and functional groups to template molecules, which make MIP recognize template from its analogues [9,10]. Due to the advantages of easy preparation, low cost, predictable specific recognition, and high stability, MIP has been extensively applied in many fields, especially in electrochemical sensing [11–14]. As a recognition element, MIP is often modified on electrode surface to prepare an electrochemical sensor for template molecule selective detection.

To overcome the non-specific adsorption, the conductivity of MIP

layer on electrode surface usually is poor. However, this in turn limits the sensitivity of the sensor [15]. Thus, many efforts have been made to improve the sensitivity of MIP based electrochemical sensors. The most common approach is the modification of electrode with conductive materials, such as noble metals nanoparticles [16,17], carbon nanotubes [18,19], graphene [20,21], 3D multiporous materials [22,23], and conductive polymers [24]. For electro-active template, current of template increases with the adsorption of template, which is directly used for template detection [25,26]. For non electro-active template, electro-active probes have been employed for template detection [27,28]. Electro-active probe can reach electrode surface through the imprinted cavities and then is oxidized/reduced. With the adsorption of template in imprinted cavities, the current of probe decreases since the adsorbed template blocks the permeation of the probe. Thus, the decrease of probe current can be used to detect non electro-active template.

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In recent years, ratiometric strategy also has been applied in the electrochemical sensing. A consistent current of electro-active probe is divided by analyte's current to obtain a ratiometric signal, which not only can be used for analyte sensitive detection but also overcome the potential interferences from intrinsic background [29,30]. Tian's group used 6-(ferrocenyl) hexanethiol as the probe to accurately detect Cd^{2+} , Cu^{2+} , and pH value with good sensitivity [31–33]. Ferrocene and methylene blue were added in electrolyte solution directly for ratiometric signal based electrochemical sensors fabrication by Wang's group [34,35]. On the other hand, both current signals of electro-active probe and analyte would change with the increase of analyte's concentration [36]. Our group used Prussian blue (PB) as the electro-active probe, which was immobilized on electrode surface to fabricate a MIP based electrochemical sensor. The imprinted cavities in MIP film provided many electronic transfer channels for PB redox. Thus, increased current of template and decreased current of PB were combined as the ratiometric signal for template sensitive detection [37].

Besides PB, thionine (TH) also has been widely used as an electro-active probe in the field of electroanalytical chemistry based on the constantly changed or unchanged current of TH [38–41]. A label-free microfluidic paper-based sensor was prepared by modifying gold nanoparticles/reduced graphene oxide/TH nanocomposites on electrode surface. As the electrochemical mediator, the increase of TH current was proportional with the increase of concentration of prostate specific antigen [42]. P-Benzaldehyde-thionine amine-aldehyde condensation reaction was carried out to prepared electro-active Schiff base polymers, which were modified on electrode to fabricate an electrochemical sensor. The peak current of TH remained unchanged, which was used as the invariant reference signal for glucose ratiometric detection [43]. In recent years, TH also has been used as the electro-active probe to be immobilized on electrode surface for imprinted electrochemical sensor preparation [44]. An electrochemical sensor was prepared by successive modification of carbon nanotubes, polythionine, and MIP on electrode surface for propyl gallate ratiometric detection based on the increase of propyl gallate current and the decrease of TH current [45].

In this work, carbon nanoparticles (CNPs), simply prepared by the collection of candle ashes, were modified on glassy carbon electrode (GCE). CNPs were immobilized by chitosan due to its excellent film-forming ability of high water permeability, good adhesion, and susceptibility to chemical modifications [46–48]. Then, a MIP based electrochemical sensor was fabricated by the electropolymerization of pyrrole in the presence of aloe-emodin (AE, template) on CNPs modified electrode. Electro-active TH can be adsorbed into imprinted cavities since it has very similar structure with AE. It was subsequently replaced by AE due to the complementation of the imprinted cavities to the size, shape, and functionality of the template. Thus, (i) CNPs improved the sensitivity of the sensor; (ii) the replacement of TH by AE, demonstrated the good selectivity of the sensor and provided the second current signal; (iii) the replacement process resulted in the decrease of TH current and the increase of AE current, which were combined as a ratiometric signal for AE sensitive detection. Therefore, the ratiometric strategy based sensor has achieved the sensitive and selective detection of AE at the same time.

2. Experimental

2.1. Chemicals

Pyrrole and thionine (TH) were purchased from Sigma-Aldrich Co., LLC. Aloe-emodin (AE), quercetin (QR), puerarin (PU), rutin (RU), emodin (EM), and rhein (RH) were ordered from Aladdin (Aladdin, China). Chitosan, Lithium perchlorate (LiAlO_4), sulfuric acid (H_2SO_4) were purchased from Sinopharm Chemical Reagent Co. All other reagents were of at least analytical-reagent grade. And double-distilled deionized water was used for all solutions.

2.2. Apparatus

Cyclic voltammetry (CV) and differential pulse voltammetry (DPV) were performed on CHI 660C workstation (Chenhua Instruments Co., Shanghai, China) with a conventional three-electrode system. A bare or modified glassy carbon electrode (GCE) was served as a working electrode. A saturated calomel electrode and a platinum wire electrode were used as a reference electrode and a counter-electrode, respectively. Field emission scanning electron microscope (FE-SEM) images were obtained on an S-4800 field emission scanning electron microscopy (Hitachi, Japan). Elemental analyses of C, H, and N were performed on an Elementar Vario MICRO Elemental Analyzer. Transmission electron microscopy (TEM) images were obtained on a JEM-2100 instrument at an accelerating voltage of 100 kV. Fourier transform infrared (FTIR) spectra were obtained on a Shimadzu FTIR-8400S spectrometer by pressing a small amount of sample and KBr crystal into a plate.

2.3. Computational simulation

Computational works in this paper were performed using Gaussian 03 suite of programs. Geometries of all investigated chemical systems, including pyrrole, AE, TH and the complexes of pyrrole with AE or TH, were fully optimized without symmetry constraints by employing Becke's three-parameter hybrid functional B3LYP using 6-311 + G(d, p) basis set. All optimized structures were confirmed to be real minima by frequency calculation at the same level of theory (no imaginary frequency). Unscaled zero-point energies (ZPE) were abstracted from frequency analysis to make thermochemical correction to electronic energies of all studied species. The zero-point corrected energies of the most stable conformations of AE and TH were used to calculate the binding energies (ΔE) between pyrrole and AE or TH according to Eq. (1) [49,50]:

$$\Delta E = E_{(\text{pyrrole-AE or pyrrole-TH complex})} - E_{(\text{pyrrole})} - E_{(\text{AE or TH})} \quad (1)$$

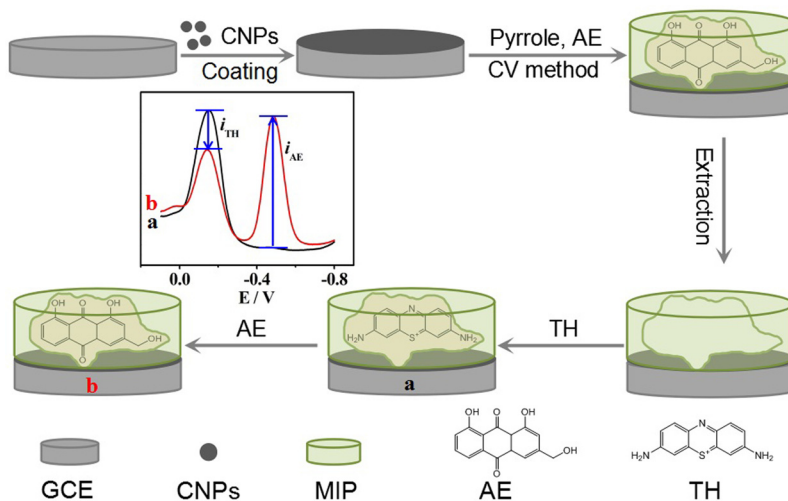
2.4. Preparation of the composite of CNPs and chitosan

The CNPs were prepared according to the literatures with slight modification [51,52]. Briefly, a beaker was set on the top of a smoldering candles to get candle soot. Then 20 mg collected candle soot was suspended in 20 mL water and ethanol mixed solution (v/v 1:1) under the ultrasonication. The suspension was centrifuged with 3000 rpm for 4 min to remove large-size particles. The supernatant was centrifuged with 6000 rpm for 15 min. And the obtained precipitate was dried for the further use.

5.0 mg/mL chitosan solution prepared by dissolving 5 mg of chitosan in 0.05 mol/L HCl under stirring for 12 h at 85 °C. Then the pH value of the solution was adjusted to 5.0 by the addition of 0.5 mol/L NaOH solution. The prepared CNPs and the chitosan solution were added into phosphate buffer solution (PBS, 0.1 mol/L, pH 7.4) to get a homogeneous suspension under the ultrasonication with the final concentrations of 0.40 mg/mL CNPs and 0.15 mg/mL chitosan, respectively.

2.5. Fabrication of CNPs modified electrode (CNPs/GCE)

Prior to fabricating CNPs/GCE, a glassy carbon electrode (GCE) was successively polished with 0.3 and 0.05 μm $\alpha\text{-Al}_2\text{O}_3$ powder and rinsed subsequently with water in the ultrasonic cleaner. A 4 μL of CNPs and chitosan mixture solution was cast onto the electrode surface and dried at room temperature, which was labeled CNPs/GCE.



Scheme 1. The procedure for ratiometric electrochemical sensor preparation.

2.6. Fabrication of MIP modified CNPs/GCE (MIP/CNPs/GCE)

CV method was carried out from -0.6 V to 0.8 V for 20 cycles at a scan rate of 50 mV/s by immersing CNPs/GCE in 5.0 mL PBS (pH 5.0) containing 40 mmol/L pyrrole, 4 mmol/L AE, and 0.1 mol/L LiClO₄. Afterwards, the modified electrode was scanned from -0.4 V to 1.6 V for 20 cycles in sulfuric acid solution (0.5 mol/L) to extract the embedded AE molecules. This modified electrode was labeled as MIP/CNPs/GCE. The procedure for the construction of MIP/CNPs/GCE was illustrated in Scheme 1. As a control, a non-molecularly imprinted polymer modified CNPs/GCE (NIP/CNPs/GCE) was prepared under the same conditions except the omitting of AE.

2.7. Pretreatment of real samples

Cassia seed, capsulae aloes composite, and aloe vera gel samples obtained from supermarket were treated according to the literature [53,54]. The dried cassia seed was ground to homogeneous powder. 10 g cassia seed powder was macerated with ethanol at room temperature for three times. 0.05 g capsulae aloes composite powder or aloe vera gel was added 10 mL of methanol, which were mixed under ultrasound for 1 h. The extraction solutions were centrifuged and the supernates were collected for electrochemical detection.

2.8. Electrochemical measurements

All electrochemical measurements were carried out in PBS (0.1 mol/L, pH 5.0). The CV experiments were conducted from -0.8 V to 0.1 V at a scan rate of 100 mV/s. And the DPV was performed from 0.1 V to -0.8 V with pulse amplitude and width of 50 mV and 50 ms, respectively.

3. Results and discussion

3.1. Design principle of the sensor

Due to the similar structure with AE and the excellent electro-activity, TH was chosen as a smart molecule for AE selective and sensitive detection. Previously reported work usually immobilized the electro-active probe on electrode surface for ratiometric strategy based electrochemical sensor preparation. However, in the present work, TH was directly added into electrolyte. Without AE, TH molecules were adsorbed into imprinted cavities of MIP since it has the similar structure of AE. As shown in DPV curve in Scheme 1, an obvious reduction peak of TH at about -0.13 V can be observed (black curve), which indicated

that TH molecules have been adsorbed into imprinted cavities. Because of the specific adsorption ability of MIP to its template, TH molecules were replaced by AE molecules. Thus, a decreased current of TH was found when AE was added into electrolyte solution (red curve). This phenomenon just demonstrated the good selectivity of the prepared MIP based sensor to AE since template can effectively compete imprinted cavities from its analogue. At the same time, the adsorbed AE molecules were reduced on MIP/CNPs/GCE to show a new reduction peak at about -0.48 V. Based on the increase of reduction current of AE ($|\Delta i_{AE}|$) and the decrease of reduction current of TH ($|\Delta i_{TH}|$), a ratiometric signal, defined as $|\Delta i_{AE}|/|\Delta i_{TH}|$, was employed for AE sensitive detection. It's obvious that the involved TH played double roles in the present work. As an analogue of AE, TH was replaced by AE to show the decrease current of itself, which demonstrated the selective recognition ability of MIP/CNPs/GCE to AE. As an electro-active probe, the decrease TH current, caused by the addition of AE, can be combined with the increase of AE current to design a ratiometric signal for AE sensitive detection.

To predict the feasibility of the replacement process, the hydrogen binding energies between pyrrole and AE or TH (i.e. pyrrole-AE and pyrrole-TH) were theoretically investigated. The optimized geometries of pyrrole-AE and pyrrole-TH were shown in Fig. S1. The calculated energies of pyrrole, AE and TH, the binding energies of pyrrole-AE and pyrrole-TH were listed in Table S1. It's obvious that the $\Delta E_{\text{(pyrrole-AE)}} < \Delta E_{\text{(pyrrole-TH)}}$, which indicated that the adsorbed TH can be expected to be replaced by AE [55,56].

3.2. Characterization of the sensor

The morphology of the CNPs was investigated by scanning electron microscope (SEM) and transmission electron microscopy (TEM), as shown in Fig. 1A and B. It's obvious that CNPs were spherical-like particles with the average size of 40 nm, which was consistent with the literatures [51,52]. The main element of the prepared CNPs was analyzed to be carbon (elemental analysis: C 92.88%, H 1.54%, N 0.07%, O (calculated) 5.51%), which was similar with the reported results [57]. After dropping the composite of CNPs and chitosan on GCE surface, a film-like structure with the entrapping of CNPs can be observed, as shown in Fig. 1C. With the electropolymerization of MIP on CNPs/GCE, a layer of film can be observed on CNPs and the shape of the modified electrode became obscure (Fig. 1D), indicating the successful preparation of MIP/CNPs/GCE. The modified materials have been characterized by Fourier Transform infrared spectroscopy (Fig. S2), which also demonstrated the successful preparation of MIP/CNPs/GCE.

Cyclic voltammetry (CV) method was used to investigate the

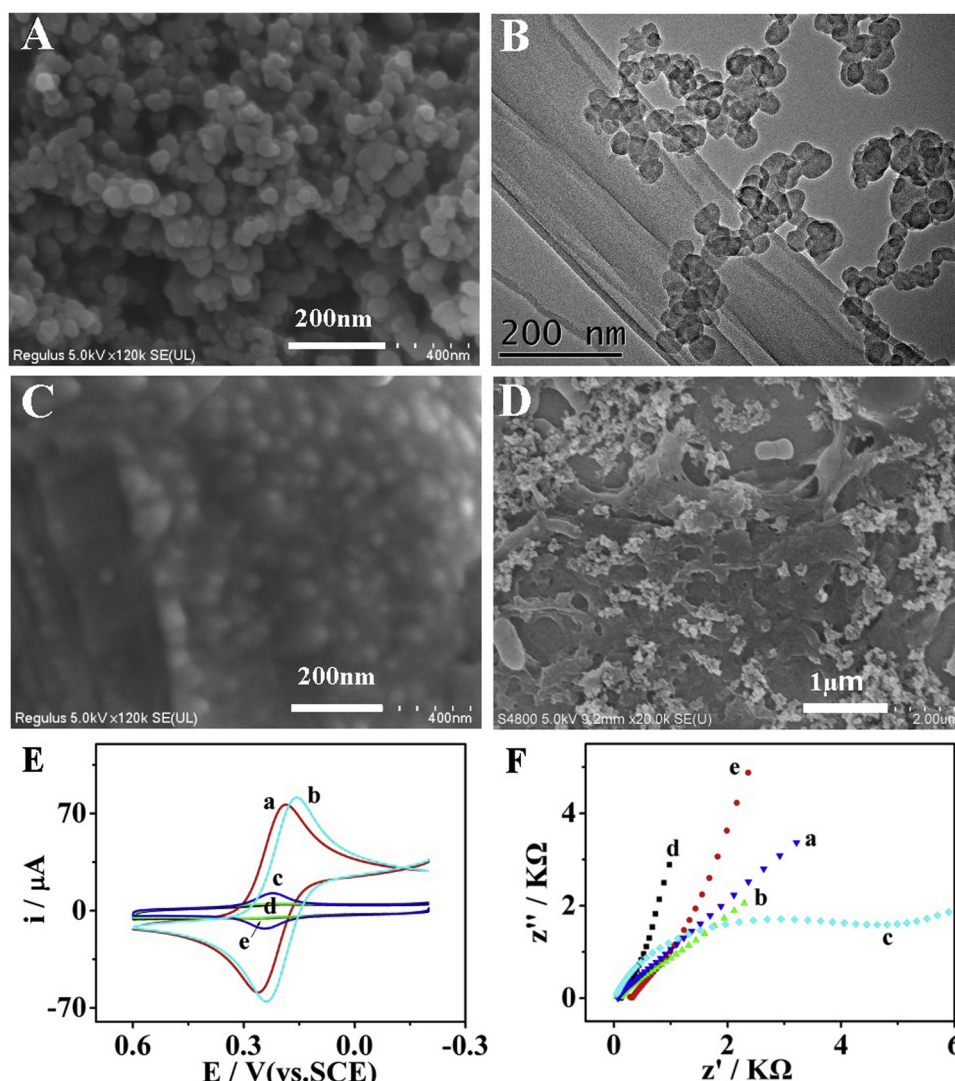


Fig. 1. SEM images of CNPs (A), CNPs/GCE (C), and MIP/CNPs/GCE (D). (B) TEM image of CNPs. (E) CV curves of bare GCE (a), CNPs/GCE (b), MIP/CNPs/GCE (c) and NIPs/CNPs/GCE (d), and MIP/CNPs/GCE rebinding 5.0×10^{-5} mol/L AE (e, almost covered by d) in 5 mmol/L $\text{Fe}(\text{CN})_6^{3-}/\text{Fe}(\text{CN})_6^{4-}$. (F) EIS curves of bare GCE (a), CNPs/GCE (b), MIP/CNPs/GCE before (c) and after (d) the extraction of AE and MIP/CNPs/GCE rebinding 5.0×10^{-5} mol/L AE (e).

properties of different modified electrodes, as shown in Fig. 1E. Compared with bare GCE (Fig. 1Ea), the peak current of $\text{Fe}(\text{CN})_6^{4-}/\text{Fe}(\text{CN})_6^{3-}$ increased when CNPs/GCE was used as working electrode (Fig. 1Eb), which can be attributed to the good conductivity of CNPs. Compared with bare GCE, an obvious increase current can be observed when CNPs/GCE was used as working electrode in 1×10^{-5} mol/L AE (Fig. S3), indicating the good conductivity of CNPs for AE. The current of $\text{Fe}(\text{CN})_6^{4-}/\text{Fe}(\text{CN})_6^{3-}$ remarkably decreased with the electro-polymerization of MIP (Fig. 1Ec) or NIP (Fig. 1Ed) on CNPs/GCE due to the poor conductivity of the polymer film. With the extraction of AE, an obvious pair of redox peak of $\text{Fe}(\text{CN})_6^{4-}/\text{Fe}(\text{CN})_6^{3-}$ can be found because the imprinted cavities in MIP film provided the electronic transfer channels for $\text{Fe}(\text{CN})_6^{4-}/\text{Fe}(\text{CN})_6^{3-}$ (Fig. 1Ec). However, the current of $\text{Fe}(\text{CN})_6^{4-}/\text{Fe}(\text{CN})_6^{3-}$ decreased with the addition of AE (Fig. 1Ee), which can be explained that the rebound AE into the imprinted cavities hindered the electronic transfer from $\text{Fe}(\text{CN})_6^{4-}/\text{Fe}(\text{CN})_6^{3-}$ to the electrode surface [58].

Electrochemical impedance spectroscopy (EIS) is an effective approach for the characterization of the modified electrode at each step. Almost the same slope of the line part of CNPs/GCE (Fig. 1Fb) and the bare GCE (Fig. 1Fa) can be observed, which can be ascribed to the good conductivity of CNPs. However, due to the poor conductivity of the

polymer film, the prepared MIP/CNPs/GCE displayed an obvious interfacial resistance before the extraction of AE (Fig. 1Fc). With the removal of AE, the resistance decreased (Fig. 1Fd). When AE was added into solution, increased resistance can be observed (Fig. 1Fe), indicating that AE has been rebound into imprinted cavities to hinder the electron transfer. The results of SEM, CV, and EIS demonstrated the successful preparation of MIP/CNPs/GCE.

CV method was employed to investigate the specific adsorption of the prepared sensor. As shown in Fig. 2A, no redox peak of AE can be found when MIP/CNPs/GCE was used as working electrode in blank solution (Fig. 2Aa), indicating the successful extraction of template molecule from the MIP film. With the addition of 5.0×10^{-5} mol/L AE, an obvious pair of redox peaks at about -0.48 V and -0.55 V can be observed, which implied that the AE molecules have been rebound into imprinted cavities. However, no redox peak can be found on NIP/CNPs/GCE either in the absence (Fig. 2Ac) or presence (Fig. 2Ad) of AE, which can be ascribed to no imprinted cavities in the NIP film.

When the modified electrode was immersed in 3.0×10^{-6} mol/L TH, a pair of redox peaks at about -0.10 V and -0.18 V can be observed on MIP/CNPs/GCE (Fig. 2Ba) and MIP/GCE (Fig. 2Bc), which corresponded to the oxidation peak and reduction peak of TH. Compared with MIP/GCE, MIP/CNPs/GCE showed much higher current of

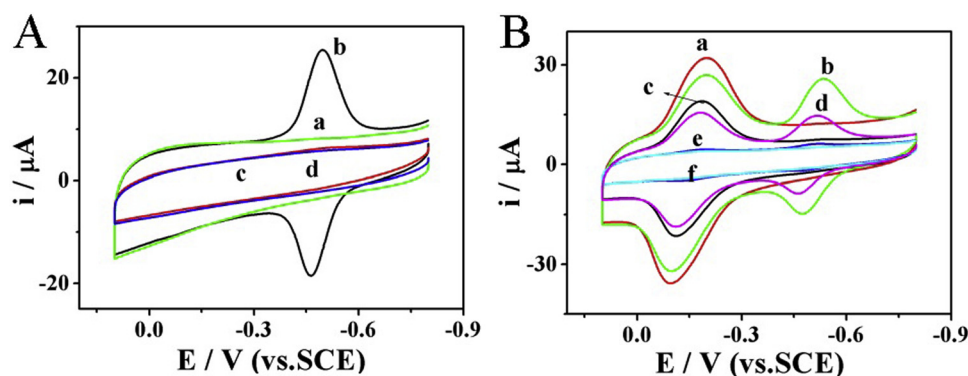


Fig. 2. (A) CV curves of MIP/CNPs/GCE (a) and NIP/CNPs/GCE (c) in blank electrolyte; CV curves of MIP/CNPs/GCE (b) and NIP/CNPs/GCE (d) in 5.0×10^{-5} mol/L AE. (B) CV curves of MIP/CNPs/GCE (a), MIP/GCE (c), and NIP/CNPs/GCE (e) in 3.0×10^{-6} mol/L TH; CV curves of MIP/CNPs/GCE (b), MIP/GCE (d), and NIP/CNPs/GCE (f) in the solution containing 3.0×10^{-6} mol/L TH and 5.0×10^{-5} mol/L AE.

TH, indicating that the modified CNPs can effectively improve the sensitivity of the sensor. With the addition of AE, the redox peaks of AE appeared on MIP/CNPs/GCE (Fig. 2Bb) and MIP/GCE (Fig. 2Bd) due to the adsorption of AE into imprinted cavities. Just because of this, the adsorbed TH in imprinted cavities was replaced by AE to cause the decrease of TH current, which well demonstrated the selectivity of MIP/CNPs/GCE to AE. On the other hand, the ratiometric strategy base on the increase of AE current and the decrease of TH current was proposed for AE detection. As for NIP/CNPs/GCE, also no obvious redox peak can be found either in TH solution (Fig. 2Be) or in the mixed solution of TH and AE (Fig. 2Bf) since no imprinted cavities were produced during the electropolymerization.

3.3. Selectivity of the sensor

Selective recognition of template molecules is an important property of MIP based sensor. Although, the replacement of TH by AE has already demonstrated the good selectivity of the prepared sensor, other analogues or co-existed compounds including quercetin (QR), puerarin (PU), rutin (RU), emodin (EM), and rhein (RH) were chosen as interferences to investigate the recognition capacity of MIP/CNPs/GCE. The structural formulas of AE and interferences can be found in Fig. S4. All of these interferences are electro-activity and their redox peaks potentials are listed in Table S2. Except for A and B, the redox peaks potentials of c, d, and e were not in the potential range of AE detection in this work.

The selectivity experiments were carried out in the mixed solution of AE and each interference with the same concentration of 5.0×10^{-5} mol/L. The current ratio ($|\Delta i_{AE+Interference}|/|\Delta i_{TH}|$) was divided by another current ratio ($|\Delta i_{AE}|/|\Delta i_{TH}|$) (Fig. 3A), which was based on the currents recorded on MIP/CNPs/GCE. When no TH was involved, $|\Delta i_{AE+Interference}|$ was directly divided by $|\Delta i_{AE}|$ under the same conditions (Fig. 3B). For QR, PU, RH, EM, and RU, MIP/CNPs/GCE displayed much smaller influence when TH was involved, indicating the good selectivity of the sensor based on the ratiometric strategy. It's worth mentioning

that EM and RH caused non-negligible influence, which can be attributed to their too similar structures (Fig. S4). Although MIP/CNPs/GCE was still used as working electrode, much obvious influence can be observed when no TH was involved (Fig. 3B), implying the ratiometric strategy endowed the sensor with better selectivity.

3.4. Determination of AE by ratiometric signal

The conditions for sensor preparation and the pH value for AE detection have been optimized to obtain the best performance of the sensor. DPV method was employed for optimization experiments, which was carried out in 5.0×10^{-5} mol/L AE solution. As shown in Fig. S5, the optimal conditions of the molar ratio (AE : pyrrole), the polymerization cycles and scan rate were 1:10, 20, and 50 mV/s, respectively. The optimized pH value of prepolymerization solution was 5.0.

Under the optimized conditions, the sensor was immersed in 3.0×10^{-6} mol/L TH solution with the successive addition of AE. As shown in Fig. 4A, the current of AE increased with the continuous decrease of TH current due to the replacement of TH by AE. The ratiometric signal ($|\Delta i_{AE}|/|\Delta i_{TH}|$) was proportional to the concentration of AE in two ranges of 1.0×10^{-7} – 1.0×10^{-6} mol/L and 5.0×10^{-6} – 1.0×10^{-4} mol/L (Fig. 4B). The linear regression equations were $|\Delta i_{AE}|/|\Delta i_{TH}| = 0.12677 + 0.33633 c_{AE}$ (μM) and $|\Delta i_{AE}|/|\Delta i_{TH}| = 0.89652 + 0.00897 c_{AE}$ (μM), respectively. A limit of detection (LOD) (S/N = 3) for AE was measured and calculated to be 7.5×10^{-8} mol/L. For comparison, the sensor was also used for AE direct detection without the addition of TH. It was found that the linear range and LOD were 1.0×10^{-7} – 1.0×10^{-6} mol/L and 5.0×10^{-6} – 1.0×10^{-5} mol/L, 9.8×10^{-8} mol/L, respectively. The results demonstrated that the proposed ratiometric strategy can efficiently improve the sensitivity for AE detection. Compared with other reported methods for AE detection (Table S3), the present electrochemical sensor showed wider linear range or lower LOD for AE detection.

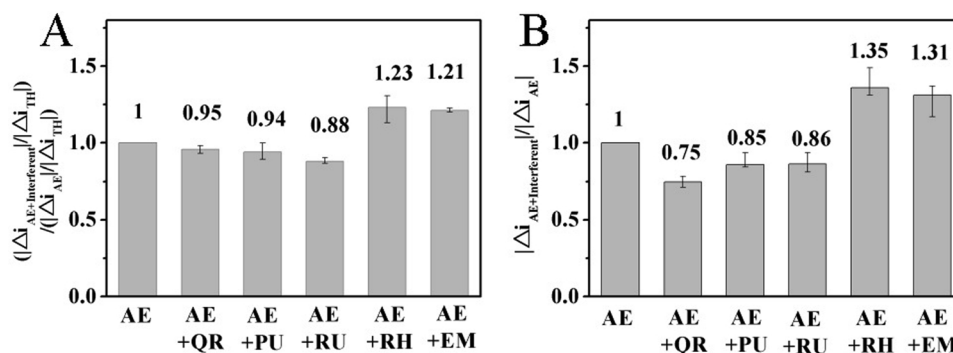


Fig. 3. The sensing performance of MIP/CNPs/GCE in different solution (AE, AE + QR, AE + PU, AE + RU, AE + RH, AE + EM) based on ratiometric strategy (A) and single signal (B). (Concentration of all compounds were 5.0×10^{-5} mol/L.).

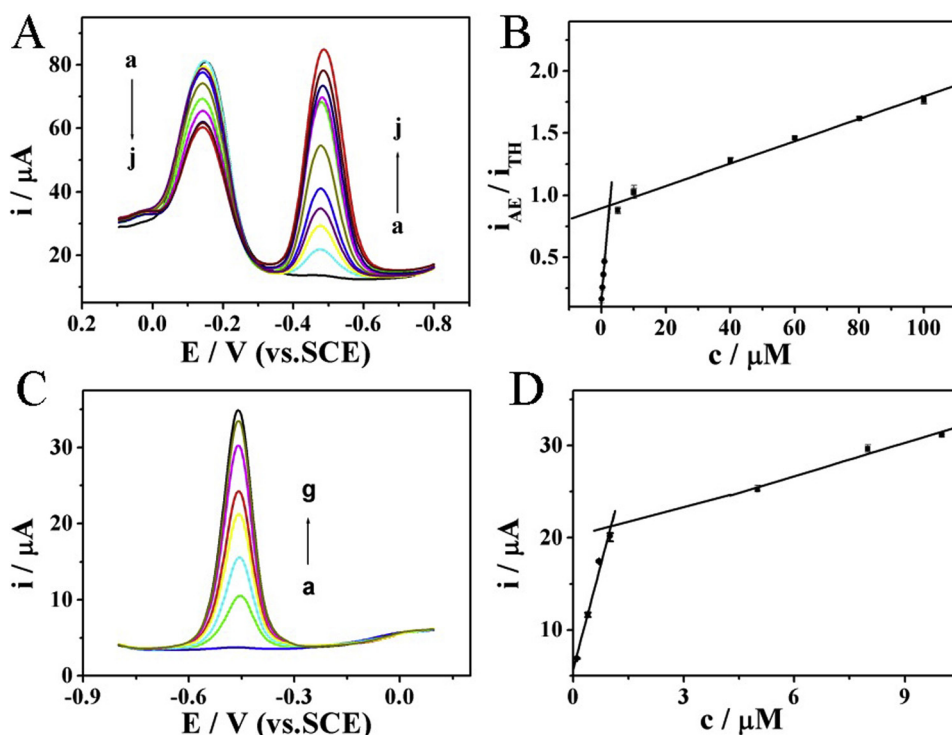


Fig. 4. (A) DPV curves of 3.0×10^{-6} mol/L TH with the successive addition of AE on MIP/CNPs/GCE (a–j: 1.0×10^{-7} , 4.0×10^{-7} , 7.0×10^{-7} , 1.0×10^{-6} , 5.0×10^{-6} , 1.0×10^{-5} , 4.0×10^{-5} , 6.0×10^{-5} , 8.0×10^{-5} , and 1.0×10^{-4} mol/L). (B) Calibration curves for AE detection based on ratiometric strategy: two linear ranges of 1.0×10^{-7} – 1.0×10^{-6} mol/L and 5.0×10^{-6} – 1.0×10^{-4} mol/L. (C) DPV curves of the successive addition of AE on MIP/CNPs/GCE (a–g: 1.0×10^{-7} , 4.0×10^{-7} , 7.0×10^{-7} , 1.0×10^{-6} , 5.0×10^{-6} , 8.0×10^{-6} , and 1.0×10^{-5} mol/L). (D) Calibration curves for AE detection based on single signal: two linear ranges of 1.0×10^{-7} – 1.0×10^{-6} mol/L and 5.0×10^{-6} – 1.0×10^{-5} mol/L.

3.5. Repeatability, reproducibility, stability, and real samples detection of the sensor

In order to investigate the real application of the sensor, repeatability, reproducibility, and stability were studied based on $|\Delta i_{AE}|/|\Delta i_{TH}|$. Five sensors, prepared under the same conditions, were used to detect 5.0×10^{-5} mol/L of AE with a relative standard deviation (RSD) of 3.35%. After five cycles of detection/extraction/adsorption, the RSD of the ratiometric signal recorded on the same sensor was 4.29%. And the sensor was used to detect AE every three days. The sensor kept 91.7% of its original current of AE with a RSD of 4.80%.

Due to the good repeatability, reproducibility, and stability, the prepared sensor was employed to determine AE in cassia seed, capsulae aloes composite, aloe vera gel samples, and AE reference substance (second grade). When 5.0 mL of each prepared sample solution was monitored directly by MIP/CNPs/GCE, no AE was found in each sample. Then, the samples were added with a known concentration of AE. The results of the samples with AE spiking were summarized in Table S4. The recoveries of three samples ranged from 100.2% to 103.0%. In addition, the RSD ranged from 1.4% to 3.9%. Three kinds of redox molecules including ascorbic acid (AA), dopamine (DA), and epinephrine (EP) with the same concentration of 1.0×10^{-5} mol/L were added in each real sample, which were detected using standard addition method. As summarized in Table S5, the ranges of recovery and the RSD were 96.8%–102.4% and 1.2%–4.7%, respectively. The results indicated that the added AA, DA, and EP did not disturb the sensor for AE detection although they were electro-active molecules. Besides, a secondary reference of AE was chosen to evaluate the reliability of the sensor for AE detection. The results showed that the values were consistent with the labeled values (Table S6). With the addition of known concentrations of AE, the recoveries ranged from 96.1% to 97.5%. All the results indicated that the prepared sensor can be used as an effective and reliable sensing platform for the determination of AE in real samples.

4. Conclusions

In summary, CNPs and MIP were successively modified on GCE to

fabricate an electrochemical sensor for AE detection. The modified CNPs improved the sensitivity of the sensor. The introduced TH played double roles during the AE detection process. As the analogue, the replacement of TH in imprinted cavities by AE directly demonstrated the good selectivity of the prepared sensor. As the electro-active probe, the increase of AE current and decrease of TH current were combined as the ratiometric signal for AE sensitive detection with the linear range of 1.0×10^{-7} – 1.0×10^{-4} mol/L. Also, the sensor was used for the real samples detection with satisfactory results. The smart electro-active probe with double roles proposed here maybe open a new way for other compounds ratiometric detection with good sensitivity and selectivity. Besides electro-active template, developing ratiometric strategy based electrochemical sensor for non electro-active template selective and sensitive detection needs to be devoted in the future research.

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Appendix A. Supplementary data

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